

Day 15 – Cyber Security & Ethical Hacking

OWASP Juice Shop: Understanding Modern Web Application Failures

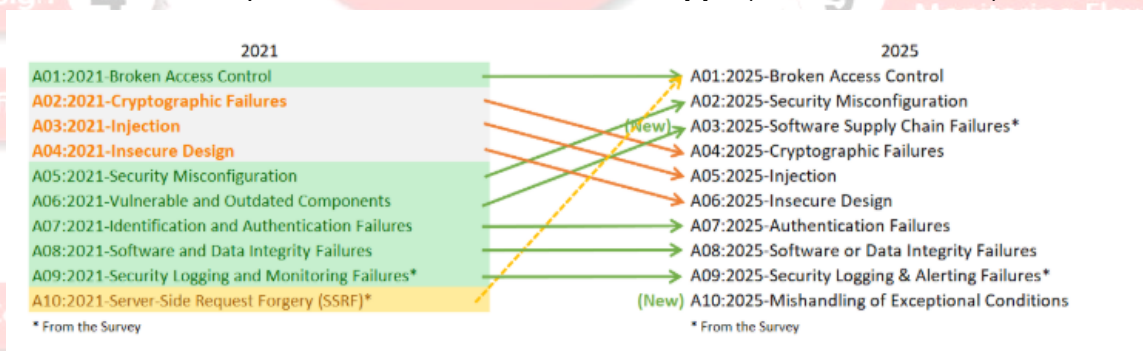
♦ Core Insight

- Modern applications rarely fail due to *simple bugs*

- They fail due to:

- Logic flaws
- Broken authentication
- Improper validation

- OWASP Juice Shop simulates **real-world modern apps** (SPA, APIs, tokens)



◆ Definition: What is OWASP Juice Shop?

- **OWASP Juice Shop** is an intentionally vulnerable **modern web application**

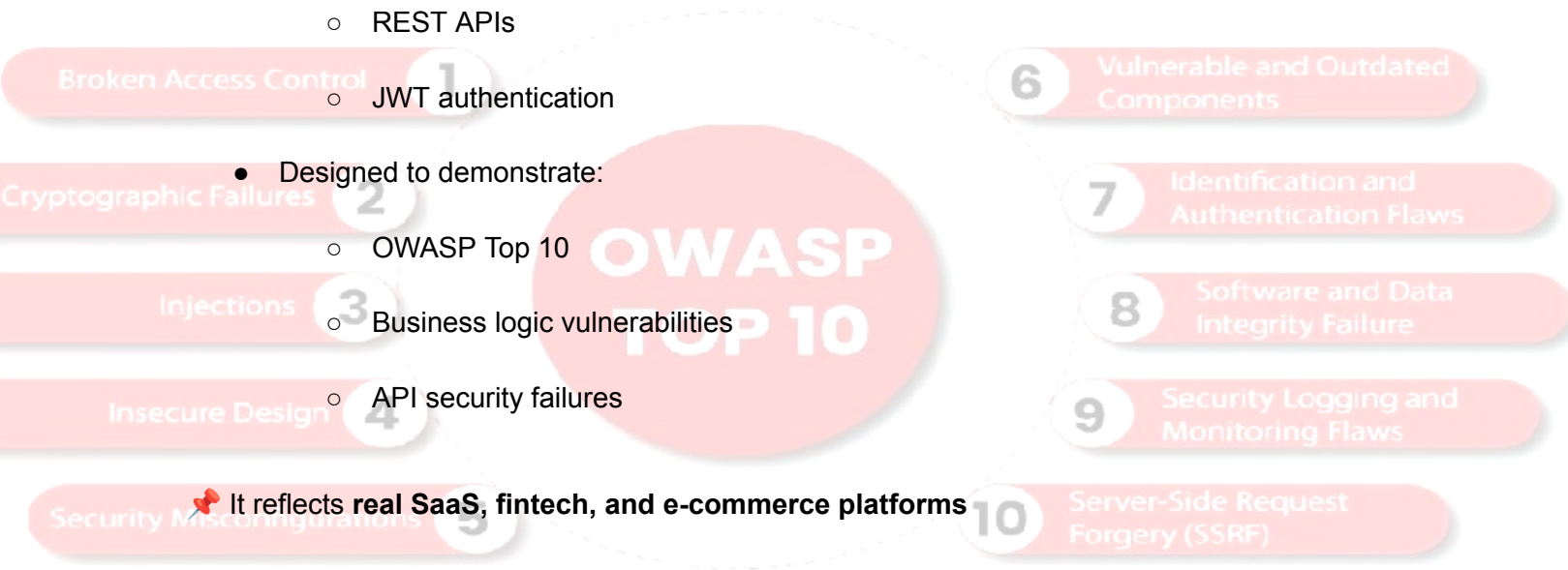
- Built using:

- JavaScript
- REST APIs
- JWT authentication

- Designed to demonstrate:

- OWASP Top 10
- Business logic vulnerabilities
- API security failures

✚ It reflects **real SaaS, fintech, and e-commerce platforms**



◆ Why Juice Shop Is Critical for Advanced Learning

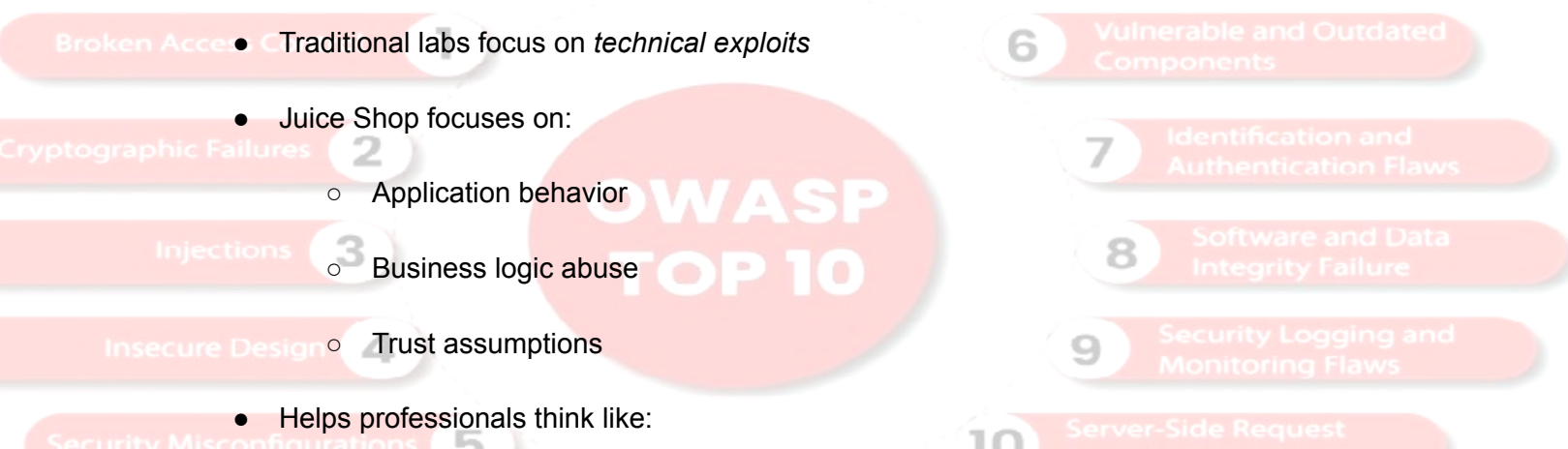
- Traditional labs focus on *technical exploits*

- Juice Shop focuses on:

- Application behavior
- Business logic abuse
- Trust assumptions

- Helps professionals think like:

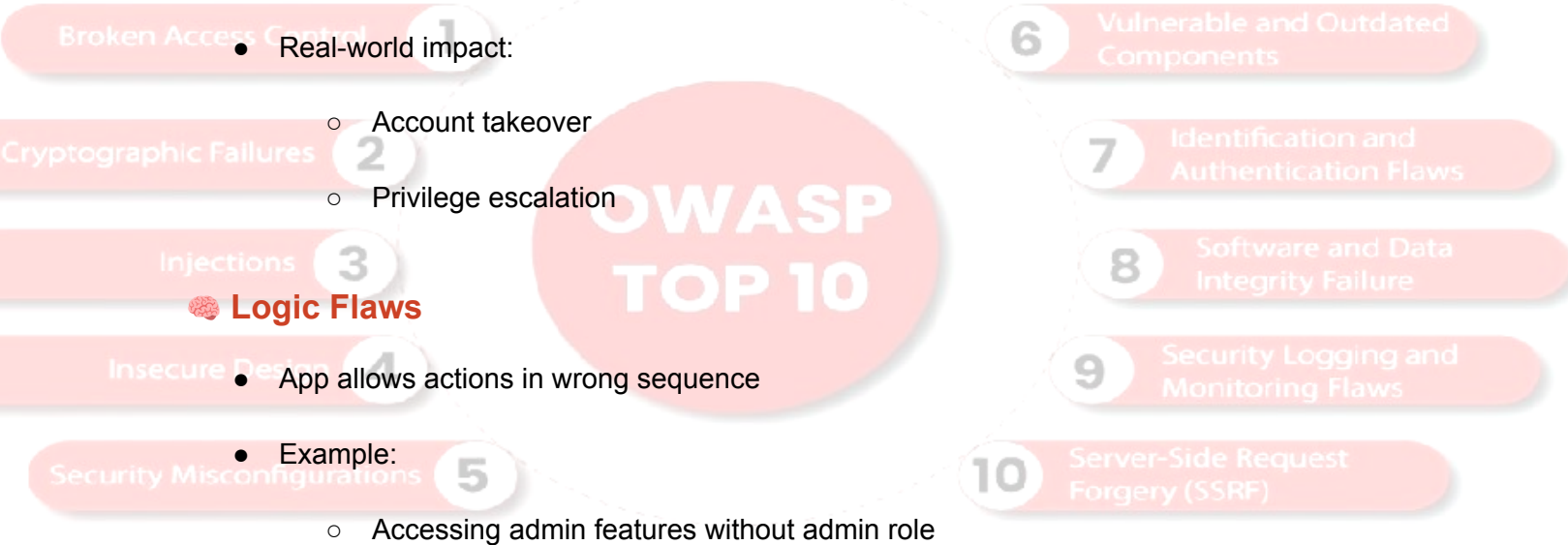
- Attackers
- Security architects
- Risk analysts



◆ Key Vulnerability Areas Explained

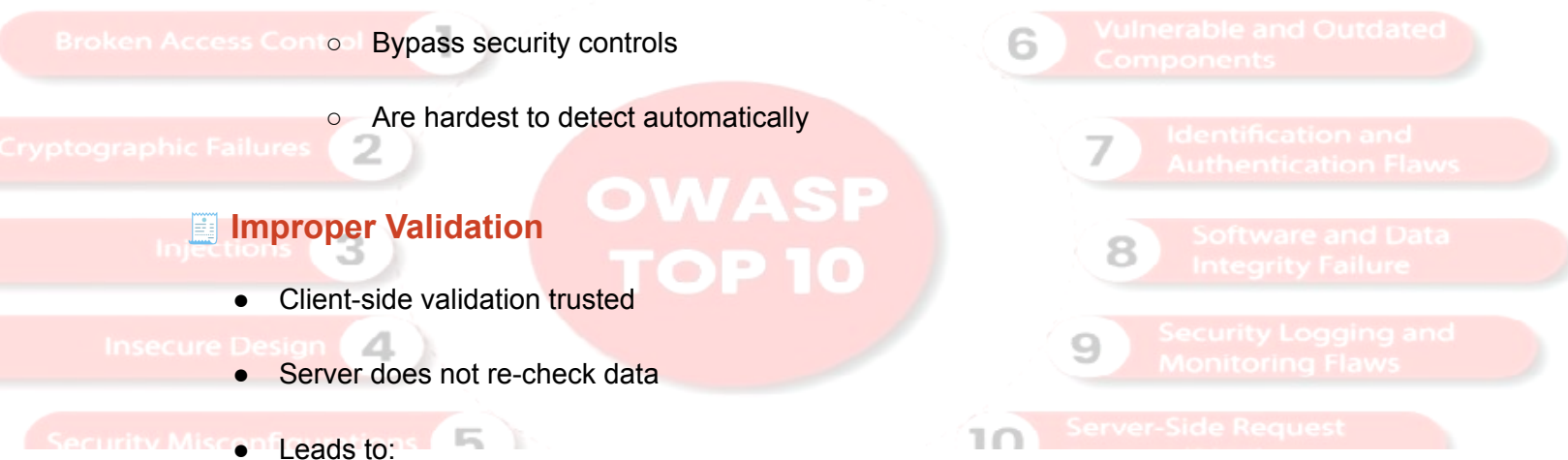
🔒 Insecure Authentication

- Weak password policies
- Token misuse
- Session handling flaws



🧠 Logic Flaws

- These flaws:

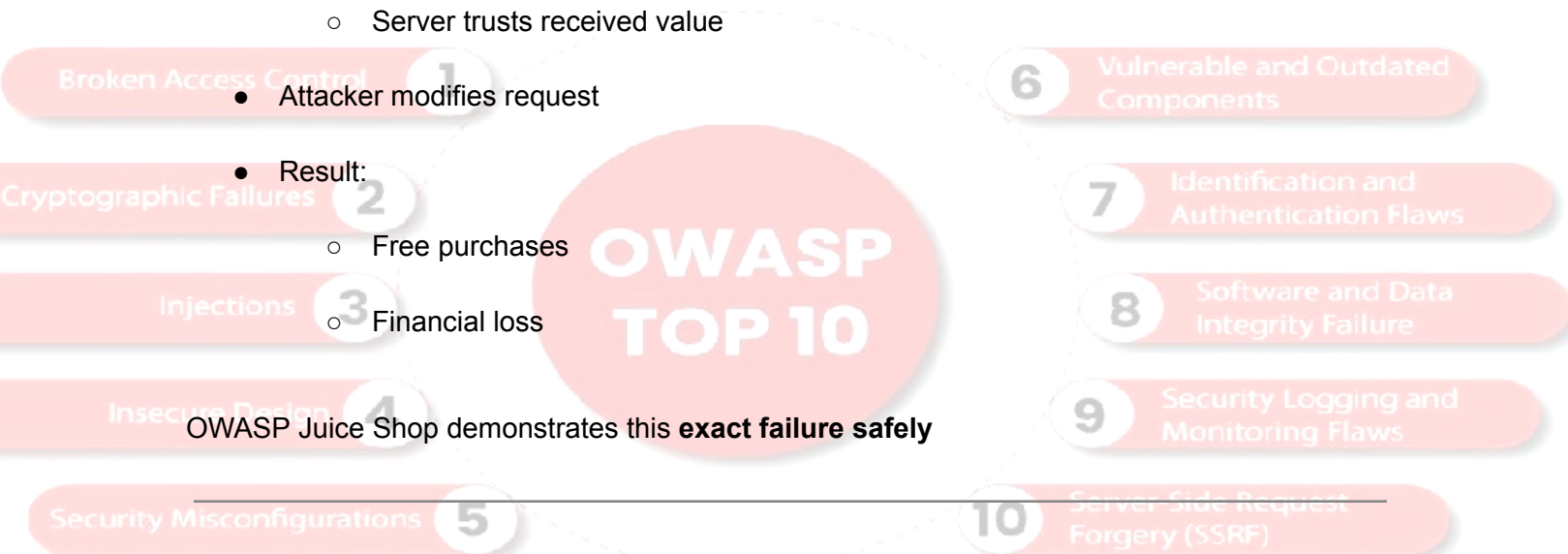


📄 Improper Validation

- Client-side validation trusted
- Server does not re-check data
- Leads to:
 - Price manipulation
 - Unauthorized actions

♦ Real-World Example Mapping

- Online shopping app:
 - Validates price only in browser
 - Server trusts received value



♦ Attacker vs Defender Perspective

• Attacker

